Experimental procedure

All experiments for the AI4Pain Challenge Dataset were conducted at the Human-Machine Interface Laboratory, University of Canberra, Australia. Participants were seated comfortably on a chair in front of a table, with both arms resting naturally on the tabletop. Physiological signals were recorded using Biosignal Plux wearable sensors and included:

- Electrodermal Activity (EDA) measured in microsiemens (µS)
- Blood Volume Pulse (BVP) measured in millivolts (mV)
- Respiration (RESP) measured in percentage of full scale or arbitrary units depending on sensor calibration
- Peripheral Oxygen Saturation (SpO_2) measured in percentage (%)

The experimental setup was designed to enable synchronised, multimodal data acquisition in a controlled and repeatable environment. All physiological data was sampled at 100 Hz.

On the right arm, the electrodes of a transcutaneous electrical nerve stimulation (TENS) machine (Medihightec Medical CO., LTD., Taiwan) were placed on the inner forearm and on the back of the hand. The location and intensity of pain stimulus were counterbalanced to avoid habituation to repeated experimental pain and to avoid confounding factors due to order effects.

At the start of each experiment, a ~60-second baseline period was recorded. This baseline period serves as the *No Pain* condition in the experimental setting. Pain stimulation was then applied at two anatomical locations (Arm and Hand). For each location, six repetitions of approximately 10 seconds were delivered at two intensity levels (*Low Pain* and *High Pain*). Therefore, each participant experienced a total of 12 repetitions for each pain intensity level (6 per location), resulting in Low Pain = 12 repetitions and High Pain = 12 repetitions. After each pain stimulus, a 40-second rest period was provided to allow physiological signals to return closer to baseline before the next stimulus was applied.

The methodology employed for data collection closely follows that outlined in our previous study [1]. A graphical representation of the two parts of the experimental procedure is presented below.

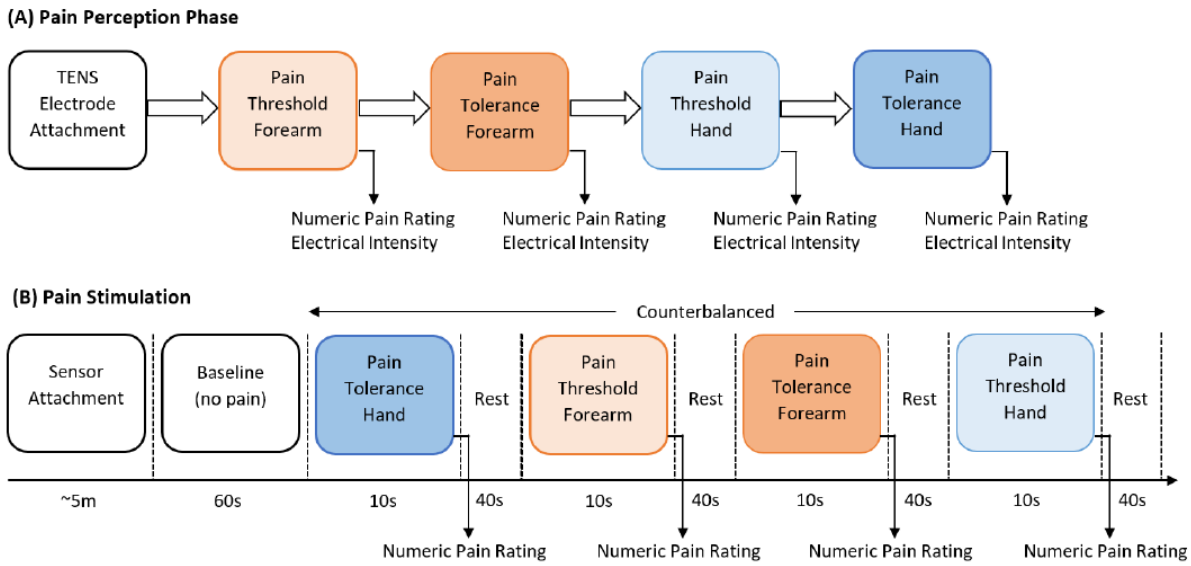

Figure 1. Graphical representation of the two parts of the experimental procedure.

AI4Pain 2026 Dataset

For the AI4Pain 2026 dataset, a balanced subset was constructed to ensure equal representation across classes. Each subject contributes 12 segments of 10 seconds for each class. The **Pain Arm** and **Pain Hand** classes correspond to stimulation periods recorded at the respective anatomical locations during the experiment. For this challenge, Low Pain and High Pain stimulation levels were combined within each location, meaning that both intensity levels are included within the same class (Pain Arm or Pain Hand). As a result, each pain class contains 12 segments of approximately 10 seconds per subject.

The **No Pain** class was constructed from baseline and rest recordings. The initial baseline recording was segmented into six non-overlapping 10-second windows, and six additional 10-second segments were obtained from the final 10 seconds of six different rest periods recorded between pain stimuli. Importantly, rest periods are not treated as a separate class and not included in this dataset, but are only used to construct additional No Pain segments to balance the dataset.

Consequently, the final dataset contains a balanced distribution across the three classes (No Pain, Pain Arm, Pain Hand), with 12 samples per class and per subject, where each sample corresponds to an approximately 10-second physiological signal segment. Each subject therefore contributes 36 segments in total.

The scientific question addressed by this challenge is whether **physiological signals alone can reliably distinguish not only the presence of pain but also the anatomical location of the stimulus across subjects**, despite inter-individual variability in physiological responses. Participants are therefore asked to develop models capable of subject-independent pain detection and localisation using multimodal physiological signals.

Participants

The dataset contains a total of 65 participants (23 Female) with a mean age of 29.06 and standard deviation of 8.28. The dataset is divided in training (41 participants), validation (12 participants), and testing (12 participants) sets. Initially, the training and validation sets are released, followed by the testing set without labels at a later stage.

[1]. Fernandez Rojas, R., Hirachan, N., Brown, N., Waddington, G., Murtagh, L., Seymour, B., & Goecke, R. (2023). Multimodal physiological sensing for the assessment of acute pain. *Frontiers in Pain Research*, 4, 1150264.